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EXECUTIVE REPORT

CAPSTONE PROJECT – GRC208

April 22, 2026

PROBLEM AND OBJECTIVE

Problem

- Traditional compliance is manual and slow
- Cloud environments change rapidly
- Leads to compliance gaps

Objective

- Build an automated GRC platform on AWS
- Enable continuous compliance monitoring
- Improve visibility and audit readiness

Manual Compliance

Periodic Checks
Delayed detection
Human error risk

Automated GRC Platform

Continuous monitoring
Real-time alerts
Consistent enforcement

MULTI-TIER SYSTEM ARCHITECTURE

- Network Layer (VPC, Subnets)
- Data Layer (RDS, S3, DynamoDB)
- Application Layer (Lambda, ECS)
- Security & Monitoring (Config, CloudTrail, IAM)



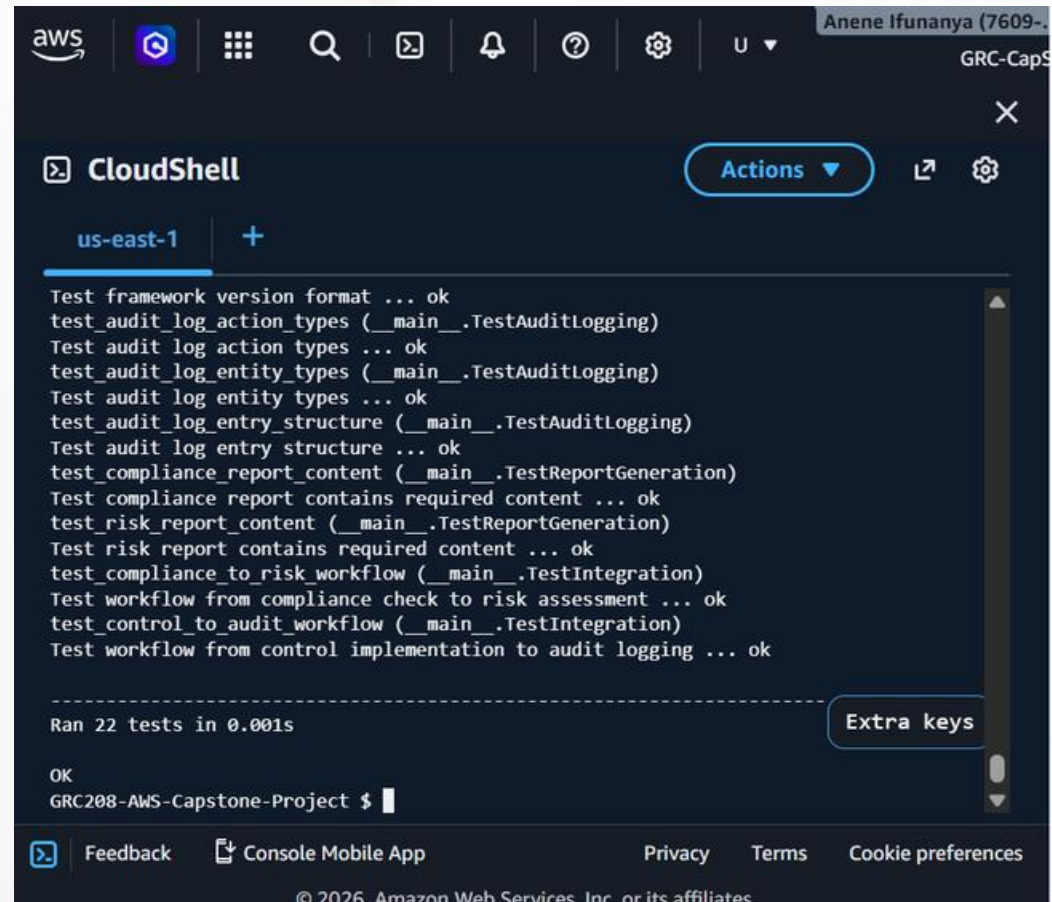
KEY IMPLEMENTATION



- CloudFormation – to deploy Infrastructure as Code
- AWS Config – to continuously monitor 500+ resources
- Lambda – to automate compliance checks
- EventBridge – to trigger checks hourly
- CloudTrail – to log all API activity

RESULTS AND VALIDATION

- 5/5 deployment phases completed
- 13+ AWS services deployed
- 22/22 tests passed
- Continuous compliance monitoring enabled
- Supports ISO, NIST, PCI DSS, HIPAA, GDPR, SOC 2



The screenshot shows the AWS CloudShell interface. The terminal output displays the results of 22 tests, all of which passed. The tests are categorized into TestAuditLogging, TestReportGeneration, and TestIntegration. The output is as follows:

```
Test framework version format ... ok
test_audit_log_action_types (__main__.TestAuditLogging)
Test audit log action types ... ok
test_audit_log_entity_types (__main__.TestAuditLogging)
Test audit log entity types ... ok
test_audit_log_entry_structure (__main__.TestAuditLogging)
Test audit log entry structure ... ok
test_compliance_report_content (__main__.TestReportGeneration)
Test compliance report contains required content ... ok
test_risk_report_content (__main__.TestReportGeneration)
Test risk report contains required content ... ok
test_compliance_to_risk_workflow (__main__.TestIntegration)
Test workflow from compliance check to risk assessment ... ok
test_control_to_audit_workflow (__main__.TestIntegration)
Test workflow from control implementation to audit logging ... ok

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Ran 22 tests in 0.001s

OK
GRC208-AWS-Capstone-Project $
```

At the bottom of the terminal window, there are links for Feedback, Console Mobile App, Privacy, Terms, and Cookie preferences. The footer indicates copyright 2026 Amazon Web Services, Inc. or its affiliates.

CHALLENGE AND CONCLUSION

Challenge

- RDS was deployed in private subnet no external access

Solution

- Used Lambda inside VPC to load data securely

Conclusion

- Built a fully automated GRC system
- Improved real-time compliance and visibility





THANK YOU

This learning opportunity
significantly strengthened
my understanding of
compliance automation
and cloud security.